

Newton's First Law of Motion

METC



BACKGROUND – EARLY IDEAS

1.

Before Isaac Newton, people such as Aristotle believed:

- Continuous force is needed to keep an object moving

2.

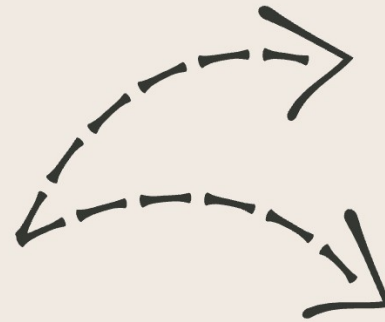
Later, Galileo Galilei proposed:

- Objects stop because of friction, not because force disappears



HOW THE LAW WAS DEVELOPED

An object will remain at rest or continue to move at a constant velocity unless acted upon by a resultant force.



Galileo's idea:
If there were no friction, objects would continue moving forever

Newton built on this and formulated a general rule about motion

Definition of Newton's First Law

Also known as: Law of Inertia

KEY TERMS

REST

The object is not moving; its velocity is zero.

CONSTANT VELOCITY

The object moves at a steady speed in a straight line (no change in speed or direction).

RESULTANT FORCE

The overall force acting on an object after all individual forces are combined.

ABOUT ISAAC NEWTON

Isaac Newton:

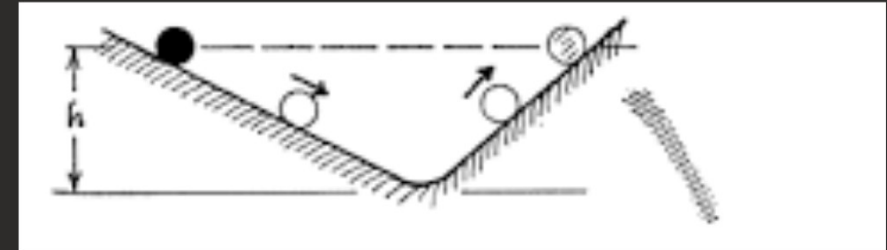
- English scientist (17th century)
- Developed the three laws of motion
- Made major contributions to physics and mathematics
- One of the founders of classical mechanics



DEMONSTRATION (IDEAL EXPERIMENT)

Galileo's Inclined Plane Thought Experiment

- A ball rolls down one slope and up another
- It always reaches the same height (if no energy loss)



<https://youtu.be/ISbqay-F-k0>

If the second
slope becomes
flatter:
The ball travels
further

Conclusion:
Without
friction, the
ball would
move forever

Real-Life Examples

Example 1: Car Braking

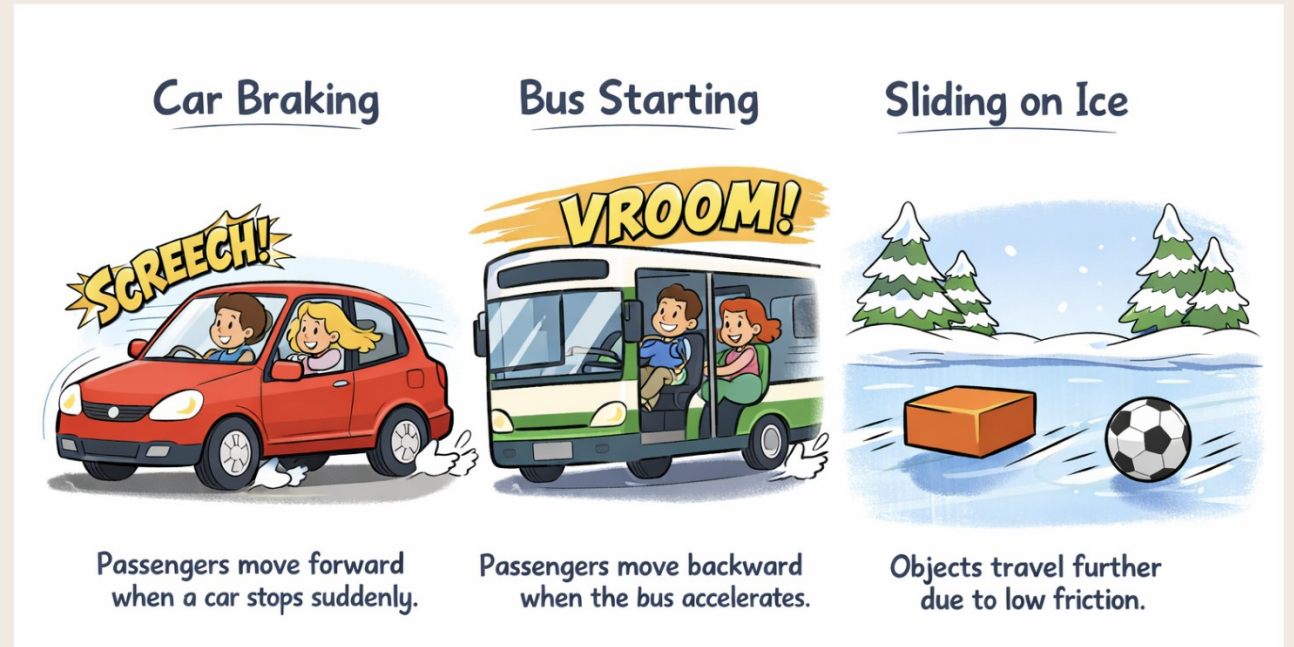
- Passengers move forward when a car stops suddenly

Example 2: Bus Starting

- Passengers move backward when the bus accelerates

Example 3: Sliding on Ice

- Objects travel further due to low friction





1. Objects resist changes in motion (inertia)
2. Motion continues unless a force acts
3. Friction is why objects stop in real life



SUMMARY



Review Questions

1. Why do objects stop moving in real life if Newton's First Law says they should keep moving?
2. What role does friction play in motion?
3. Can an object move without any force acting on it? Explain.